Indiana University
Fall 2020

SPEA-E 552
A. Ward

Name: SOLUTION + GRADE NOTES

Quiz #3 (Lectures 11-14)

$\sum$ = 34 POINTS AVAILABLE

1. For Lecture 14 you watched Dr. R. Barnes from Colorado College discuss her research. What was one element in her talk where you saw a parallel with material or concepts from our class this semester? Your response should include sufficient detail to explain your response (i.e., not just 'water').

ANSWERS WILL VARY. [+4 MAX. AVAILABLE]

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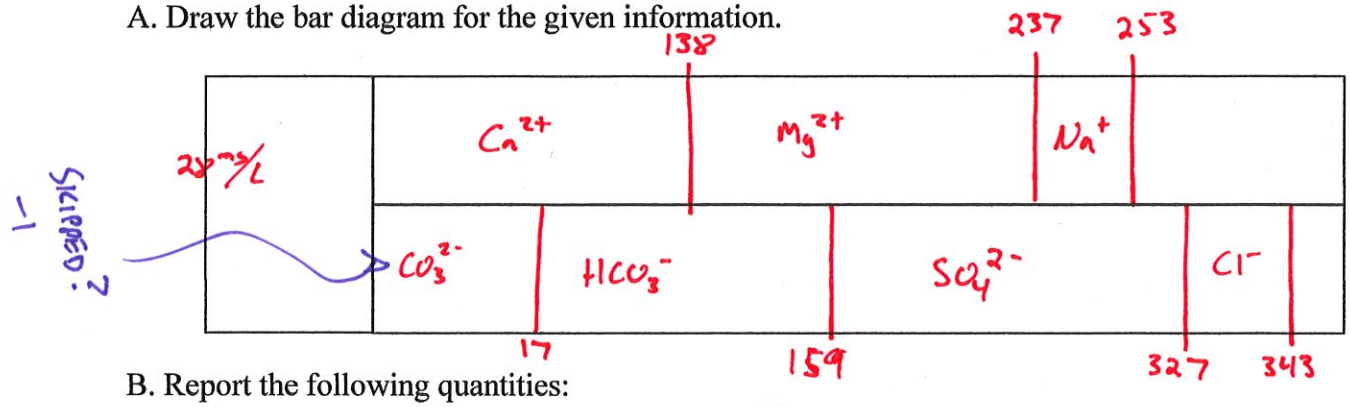
2. Softening question

Water to be softened has the following chemistry (all expressed in $CaCO_3$ equivalents):

- CO_3^{2-} = 17 mg/L ✓
- HCO_3^- = 142 mg/L ✓
- Ca^{2+} = 138 mg/L ✓
- Mg^{2+} = 99 mg/L ✓
- Na^+ = 16 mg/L ✓
- SO_4^{2-} = 168 mg/L ✓
- Cl^- = 16 mg/L ✓
- CO_2 = 28 mg/L ✓

+1 EACH × 8 #'s ON BAR CHART

A. Draw the bar diagram for the given information.



-1 SKIPPED 2

B. Report the following quantities:

Carbonate Hardness: 159

Noncarbonate Hardness: ~~[illegible]~~ 78

Total Hardness: 237

+1 EACH (CORRECT OR CONSISTENT w/ ABOVE)

B. How much lime (i.e., pure CaO) is required for the softening?

TO REMOVE CO_2: 28 mg/L +1

REMOVE CARBONATE HARDNESS: 159 mg/L +1

REMOVE NONCARB. HARDNESS = 78 mg/L +1

RECOGNIZE 3 PLACES CaO is needed

$\Sigma = 265 \frac{mg}{L}$

C. How much soda ash (i.e., pure Na_2CO_3) is required for the softening?

(NEXT PG FOR PART C)

~~NON-CARB w/ Ca^{2+} = 0~~

~~NON-CARB w/ Mg^{2+} = 237 - 159 = 78 mg/L~~

$$\frac{265 \text{ mg } CaCO_3 \text{ EQ}}{L} \times \frac{1 \text{ g}}{1000 \text{ mg}} \times \frac{1 \text{ mol } CaCO_3}{100 \text{ g } CaCO_3} \times \frac{1 \text{ mol CaO}}{1 \text{ mol } CaCO_3} \times \frac{56 \text{ g CaO}}{1 \text{ mol CaO}} \times \frac{1000 \text{ mg}}{1 \text{ g}}$$

$$= 148.4 \frac{\text{mg CaO}}{L}$$

+2 FOR $CaCO_3 \rightarrow CaO$ STOICH.

C. How much Na_2CO_3?

Remove Ca^{2+} added in prev. step for non-carb. hardness

$$\frac{78\ mg\ \text{non-carb hardness}}{L} \times \frac{1\ g}{1000\ mg} \times \frac{1\ mol\ CaCO_3}{100\ g\ CaCO_3} \times$$

$$\cdots \frac{1\ mol\ Na_2CO_3}{1\ mol\ CaCO_3} \times \frac{106\ g\ Na_2CO_3}{1\ mol\ Na_2CO_3} \times \frac{1000\ mg}{1\ g}$$

$$= \boxed{82.7\ \frac{mg\ Na_2CO_3}{L}}$$

+1 Na_2CO_3 ~~for~~ needed for Ca in NCH only

+1 used NCH value

+2 stoichiometry

3. A horizontal flow settling basin was designed to treat 50 m^3/s of discharge. The basin in 8-m wide, 2-m deep, and 50-m long. The basin was initially designed to remove suspended sediment, but it is now being evaluated for the potential to remove microplastics (i.e., tiny plastic particles; a density of 1500 kg/m^3). Assume water density is 997 kg/m^3.

A. Consider a microplastic bead with a diameter of 0.05-mm and density of 1500 kg/m^3. What fraction of these microplastics would be removed in the basin as presently designed?

OVERFLOW VELOCITY = $v_o = \frac{Q}{L \times W} = \frac{50\ m^3/s}{8m \times 50m} = 0.125\ \frac{m}{s}$

+2

SETTLING VELOCITY = $v_s = \frac{g(\rho_p - \rho_f)D^2}{18\mu} = \frac{9.81\ \frac{m}{s^2}(1500\ kg/m^3 - 997\ kg/m^3)(5\times10^{-5}\ m)^2}{18 \times 1.037\times10^{-3}\ \frac{N\cdot s}{m}}$

+2

$= 6.6\times10^{-4}\ \frac{m}{s}$

% CAPTURED = $\frac{v_s}{v_o} = \frac{6.6\times10^{-4}\ m/s}{0.125\ m/s} = 0.0053 \rightarrow \boxed{0.53\ \%}$

+2

B. What is the smallest diameter of microplastic that would be completely removed in the basin as presently designed?

100% REMOVED => $v_s = v_o$ + 2 RECOGNIZE THIS

$$0.125\ \frac{m}{s} = \frac{9.81\ \frac{m}{s^2}(1500\ kg/m^3 - 997\ kg/m^3)(D)^2}{18 \times 1.037\times10^{-3}\ \frac{N\cdot s}{m}}$$

↓ SOLVE

+ 2 WORK + SOLUTION

$D = 0.00069\ m$

$\boxed{D = 0.69\ mm}$